CAPSTONE ASSIGNMENT

Case 3: Performance Improvement of a college ERP Service.

A college ERP (Enterprice Resource Planning) system is used to manage important academic and administrative task such as student records, ~~administrative~~ admissions, attendance, fee management and result processing. During peak periods like admission time & results declaration, the number of users accessing the system increases significantly.

This heavy load leads to slow response time, delays in processing, and poor user experience. Since replacing the entire system is costly and impractical, the goal is to improve performance by optimizing the existing processor & memory system.

1) Amdahl's Law

Amdahl's law helps in identifying which part of the system should be improved to achieve maximum performance gain. It states that the overall performance improvement depends on how much of the system is affected by the enhancement.

In the ERP system, most delays occur due to memory access & repeated data retrival, so improving memory performance (cache) will give better results than only upgrading the processor.

Instruction Pipelining

Instruction pipelining is a technique where multiple instructions are executed simultaneously by dividing the execution process into stages:
Instruction Fetch (IF)
Instruction Decode (ID)
Execute
Memory Acess
Write Back
Insted of completing one instruction before starting another, pipeline allows overlapping of instructions, increasing throughput and improving performance.

Cache Memory
Cache memory is a small, high-speed memory located b/w the CPU & main memory. It stores frequently accessed data & instructions so that the CPU can access them quickly.

Since ERP systems repeatedly access similar data using cache memory significantly reduces access time & improves system performance.

2) Comparison of Improvements

Component	Impact	Limitation
processor Upgrade	Faster Execution	Expensive
pipelining	Improves throughput	May face hazards
Cache Memory	Reduces access time	limited size

It is clear that cache memory & pipelining together provide the best performance improvement at a lower cost.

Role of Cache Memory

cache memory reduces the time required to access frequently used data.

In ERP Systems:

- Students records
- Admission data
- Result data

are accessed repeatedly. Storing this data in cache minimizes repeated access to slower main memory.

Role of Pipelining

pipelining inc. instruction execution speed by allowing multiple instructions to be processed at the same time. This is useful in ERP systems where many operations are performed continuously.

4) Pipeline Hazards and their Impact

Data Hazard

Occurs when one instruction depends on the result of another instruction.

Solⁿ: data forwarding or stalling

Control Hazard

Occurs due to branch or jump instructions that change the flow of execution.

Structural Hazard
Occurs when multiple instructions compete for the same hardware resource.

4) Diagram

Instruction 1 : IF → ID → EX → MEM → WB
Instruction 2 : IF → ID → EX → MEM → WB
Instruction 3 : IF → ID → EX → MEM → WB
Instruction 4 : IF → ID → EX → MEM →

6) To improve the performance of the college ERP system

- Implement instruction pipelining to inc. execution speed.
- Use efficient cache memory to reduce memory access time
- Optimize frequently accessed data for better caching
- apply hazard handling technique to maintain pipeline efficiently.